

University of Calcutta
M.Sc. 1st Semester Examination, 2024
Computer Science
CSMC103-Advanced Computer Architecture

Time: 75 mins

Marks: 30

1. a. Comment on the desirable features of a processor, a member of any multiprocessor system.
- b. Comment on vector processor and vectorizing compiler.
- c. What are the different pipeline hazards? How do you avoid them?
- d. What is a Snooping cache?
- e. How will you justify the usage of cache memory? Express your claim in terms of the relevant parameter set.

[3+3+3+2+4]

2. a. Differentiate between Control-flow and Dataflow architectures. What are the alternative forms of Dataflow architectures? Comment on the deficiencies of DFA.
- b. How will you design the requisite reservation table for pipeline architecture from a given latency cycle?
- c. Why, in most cases, the synthesized value of MAL is different from the theoretical lower limiting value of MAL
- d. Write an algorithm to multiply 20 numerical values using a 3-cube static processor setup.
- e. Comment on the latest development of processors and associated technologies.

[3+3+3+3+3]